

LIR2032

1. Preface

The purpose of this product specification is to provide technical information for the rechargeable Lithium-ion button battery LIR2032, supplied by Guangzhou Markyn battery Co., Ltd.

2. Description and Model

2.1 Description	Rechargeable Lithium-ion button battery
2.2 Model	LIR2032

3. Specification

3.1 Rated Capacity	35mAh
3.2 Charging Voltage	4.20V
3.3 Nominal Voltage	3.7V at 0.2C mA
3.4 Constant Charging Method	Constant voltage 4.20V Constant current:17mA
3.5 Cut-off Discharge Voltage	3.00V
3.6 Max.Discharge Current	35mA
3.7 Max.Charge Current	35mA
3.8 Cycle Life	>500 cycles at 1C mA discharge
3.9 Ambient Temperature	
for Standard Charge	0°C~ 45°C
for Discharge	-20°C~ 60°C
3.10 Storage	
for within the temperature	-20°C~ 60°C
for within the humidity	≤75%
3.11 Energy Density	
Wh/L	~200
Wh/Kg	~90
3.12 Weight of Bare Cell	2.6g
3.13 Charge State Internal Impedance	<550mΩ

4.Appearance

Appearance shall be free from any remarkable scratch,flaws, rust, discoloration or electrolyte leakage(visible or by smell)

5.Standard Test condition

5.1 Environment Conditions

Unless otherwise specified,all test stated in this Product Specification are conducted within the temperature 15~25°C and the humidity 45~85%RH.

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5.2 Test Equipment

(1) Impedance meter

The impedance meter with AC 1kHz should be used

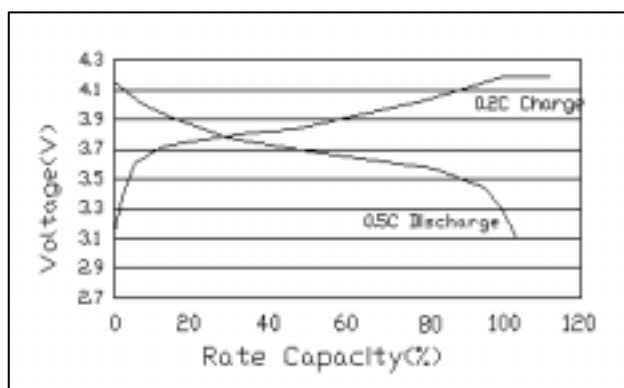
6.Test Procedure and Its Standard

Item	Measureing Procedure	Standard
6.1 Appearance	Visual	No Defect and Leak
6.2 Dimension	Caliper	As item 8
6.3 Weight	Scale	As item 3.12
6.4 Maximum Charge Current	CCCV(Constant Current Constant Voltage)	35mA
6.5 Full charge	CCCV	CC-0.2CmA CV- 4.2V total 5hrs
6.6 Open Circuit Voltage	Within 1hr after full charge,measure Open circuit voltage	>4.15V
6.7 Internal Impedance	Measure the battery with 1kHz AC	<550m Ω
6.8 Discharge Capacity	Within 1hr after full charge,discharge until final discharge,at 0.2C mA and measure the capacity	~35mAh
6.9 Maximum Discharge Current	Until final discharge voltage	35 mA
6.10 Charge/Discharge Cycle Life	Charge:CCCV,CC- 0.5CmA,CV- 4.2V for 5hr Discharge:0.5CmA to 3.00V,This charge/discharge shall be repeated 500 times	Discharge capacity should be >70% of item 6.8
6.11 Leakage Proof	After full charging,the battery shall be stored at 40 $\pm$ 2 $^{\circ}$ C and humidity 80 $\pm$ 5%for 21 days	No leakage should be observed by visual inspection
6.12 Temperature Characteristics	1)After full charge at 20 $\pm$ 5 $^{\circ}$ C ,stand at -20 $\pm$ 2 $^{\circ}$ C for 18h,then discharge at 0.2C mA and measure the capacity 2)After full charge at 20 $\pm$ 5 $^{\circ}$ C ,stand at 55 $\pm$ 2 $^{\circ}$ C for 2hrs ,then discharge at 1C mA and measure the capacity	Discharge capacity should be>60% of item 6.8 and no abnormality on its appearance and stucture
6.13 Charge Retension	After full charging,stand at 20 $\pm$ 5 $^{\circ}$ C for 28 days,measure the discharge capacity according to item 7.8	Discharge capacity should be>85% of item 6.8

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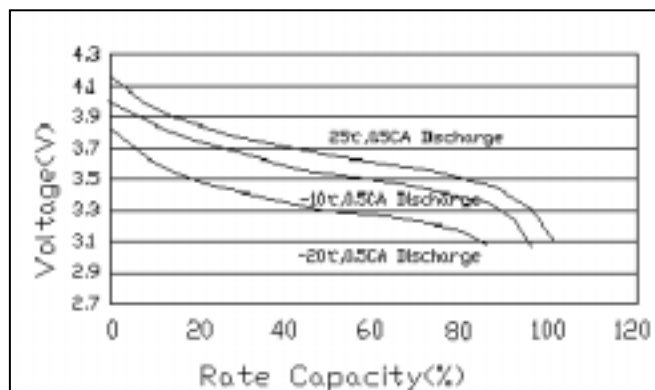
7.1 Charge/Discharge Characteristics

Charge:CC/CV 4.2V, 7mA(0.2C)*8hrs
Discharge:17mA(0.5C) Cut-off at 3.00V
Temperature:25



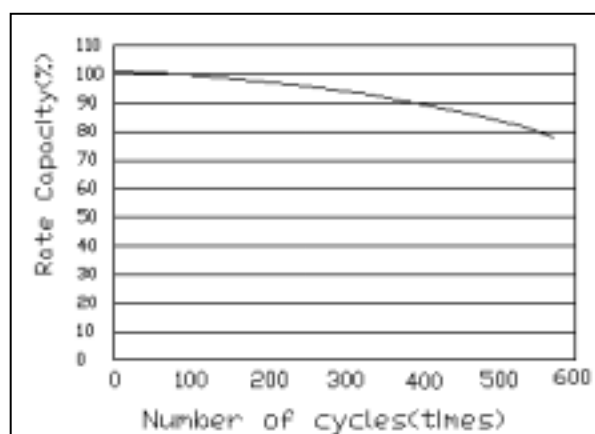
7.3 Temperature Characteristics

Charge: CC/CV 4.2V 0.2CA*8hrs
Discharge:0.5CA,Cut-off at 3.00V



7.2 Charge/Discharge Cycle Life

Charge:CC/CV 4.2V, 0.5CA*5hrs
Discharge:0.5CA,Cut-off at 3.00V
Temperature:25



8. Dimension(Bare cell) mm

